

The role of Inflammatory bowel disease in the neurological immune response to the development of Alzheimer's pathology in mice

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Research conducted on
traditional, ancestral, and
unceded territories of the
xʷməθkʷəy'əm
(Musqueam) people

2025 SBME Synergy Undergraduate Research Program



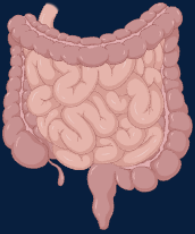
Djavad Mowafaghian
CENTRE FOR BRAIN HEALTH



genome science + technology
GRADUATE PROGRAM



Background Information



IBD: Inflammatory bowel disease

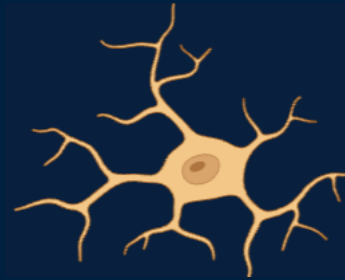
- Systemic and intestinal inflammation
- Gut microbiome dysbiosis

IBD leads to 6x higher susceptibility to AD pathology development



AD: Alzheimer's Disease

- Most common form of dementia
- Amyloid-beta plaques
- Induced by 5xFAD mutation



Microglia

- Resident immune cells
- Active and resting states
- Regulates A β plaque content
- Responds to changes in gut inflammation

Hypothesis

IBD-associated gut dysbiosis and inflammation *disrupt normal microglial function*, altering their *regulation of amyloid-beta plaque* content.

Methodology

Model Systems



Healthy-Control

IBD



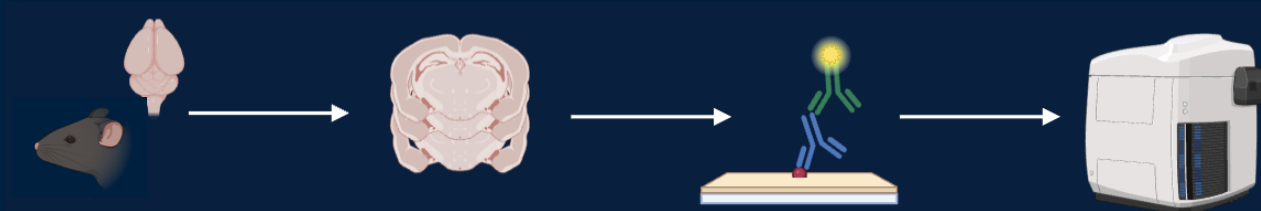
AD (5xFAD)

IBD & AD

Germ-free mice were given either *healthy* or *IBD human microbiota* from local patients

Transgenic 5xFAD engineered mice were representative **AD models**

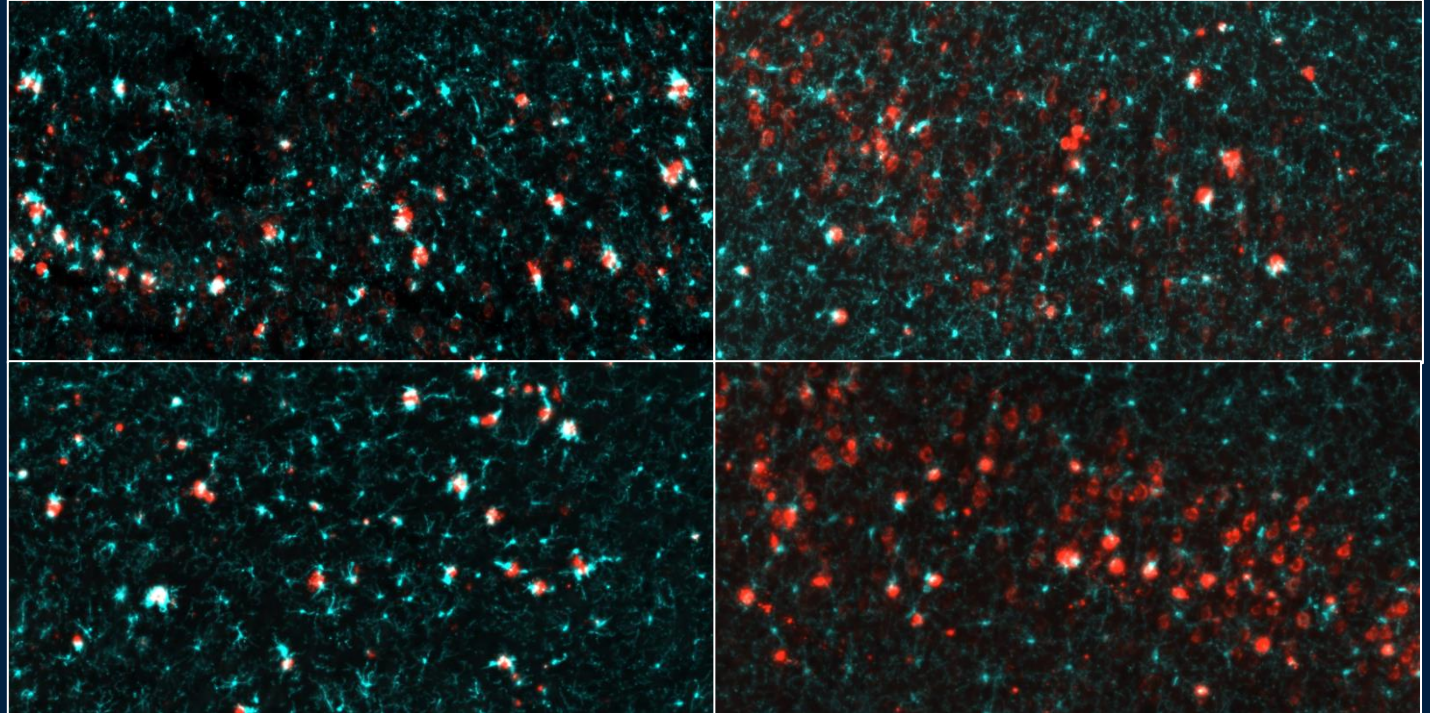
Collect Brains → Section → Immunohistochemistry → Imaging



Results & Discussion

Healthy-Control

Inflammatory bowel disease



AD
PND-60
Cortex

Legend

Red = Aβ-Plaques

Blue = Microglia

Future Directions & Implications

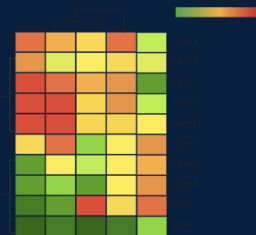
1. Quantification & Data Analysis

- Microglia
- Amyloid-beta plaques



2. Gene Expression

- Inflammatory signalling
- Amyloid & Alzheimer's genes



3. Varied Developmental Periods

- Older time periods
- Onset vs. Progression
- Establish biomarkers for IBD patients



Acknowledgements



PI: Dr. Annie V. Ciernia



PDF: Dr. Jordan E. Hamden



References



SBME Synergy Program



Ciernia Lab

